

12강

서울대학교 통계학과 이재용 교수

베이지안 통계학 선형 회귀모형





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01 선형회귀모형

01 선형회귀모형

□ 모형

$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \epsilon_i, \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2), i=1, 2, \dots, n$
 p 는 변수의 개수이고, $q = p + 1$ 이다.

□ 행렬식을 이용한 모형

$$y_{n \times 1} = X_{n \times q} \beta_{q \times 1} + \epsilon_{n \times 1}, \quad \epsilon \sim N(0, \sigma I_n)$$

$$y = (y_1, \dots, y_n)^T$$
$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$
$$\beta = (\beta_0, \beta_1, \dots, \beta_p)^T$$
$$\epsilon = (\epsilon_1, \dots, \epsilon_p)^T$$

사전분포

□ 코드에서 쓸 사전분포

$$\beta_i \sim \text{Uniform}(L_i, U_i)$$

$$\log \sigma \sim \text{Uniform}(-M, M).$$

□ 젤너-시오우

사전분포(Zellner-Siow prior)

혹은 젤너의

g -사전분포(Zellner's g -prior)

$$\tau^2 = \frac{1}{\sigma^2} \sim \text{Gamma}\left(\frac{v_0}{2}, \frac{v_0 \sigma_0^2}{2}\right)$$

$$\beta | \sigma^2 \sim N(\beta_0, g \sigma^2 (X^T X)^{-1})$$

여기서 $v_0, \sigma_0^2, \beta_0, g$ 는 상수이다.

□ 젤너-시오우 사전분포(Zellner-Siow prior)의 총조건부 사전분포

$$\beta | \tau^2, y \sim N(\tilde{\beta}, \sigma^2 \frac{g}{1+g} (X'X)^{-1})$$

$$\tau^2 | \beta, y \sim \text{Gamma}\left(\frac{v_n + q}{2}, \frac{\frac{1+g}{g}(\beta - \tilde{\beta})' X' X (\beta - \tilde{\beta}) + v_n \sigma_n^2}{2}\right)$$

$$\beta | y \sim t_{v_n}(\tilde{\beta}, \tilde{\Sigma})$$

$$\tau^2 | y \sim \text{Gamma}\left(\frac{v_n}{2}, \frac{v_n \sigma_n^2}{2}\right)$$

여기서,

$$v_n = v_0 + n$$

$$\sigma_0^2 = \frac{1}{v_n} \left[\frac{g}{1+g} \text{SSE} + \frac{1}{1+g} \text{SSE}_0 + v_0 \sigma_0^2 \right]$$

$$\tilde{\beta} = \frac{g}{1+g} \hat{\beta} + \frac{1}{1+g} \beta_0$$

$$\tilde{\Sigma} = (X'X)^{-1} X'y$$

$$\text{SSE} = \|y - x\hat{\beta}\|^2$$

$$\text{SSE}_0 = \|y - x\beta_0\|^2$$

단순선형회귀모형: 코드

```
library(rjags)
army = read.csv("army-physical.csv", header=T, sep=",")
lm.out = lm(height ~ weight, data=army)
summary(lm.out)
plot(lm.out)
modelstring="
model {
  for (i in 1:n) {
    y[i] ~ dnorm(mu[i], tau)
    mu[i] <- beta0 + beta1*x[i]
  }
  beta0 ~ dunif(L0, U0)
  beta1 ~ dunif(L1, U1)
  tau <- 1/sigma2
  log(sigma2) <- 2*log.sigma
  log(sigma) <- log.sigma
  log.sigma ~ dunif(L2, U2)
  pred70 <- beta0 + beta1*70
  pred100 <- beta0 + beta1*100
}
```

```
L0 = 0; U0 = 300
L1 = -10; U1 = 10
L2 = -100; U2 = 100
datalist = list('y'= army$height, 'x'=army$weight, 'n'=dim(army)[1],
               'L0'=L0, 'L1'=L1, 'L2'=L2, 'U0'=U0, 'U1'=U1, 'U2'=U2)
jags <- jags.model(textConnection(modelstring), data = datalist,
                  n.chains = 1, n.adapt = 100)
post <- coda.samples(jags, c('beta0', 'beta1', 'sigma', 'pred70',
                             'pred100'), n.iter=100000, thin=50)

library(dplyr)
library(coda)
library(ggmcmc)
post %>% summary
post %>% ggs %>% ggs_density
post %>% ggs %>% ggs_traceplot
post %>% ggs %>% ggs_autocorrelation
```

중회귀모형: 코드

□모형 $y_i = \beta_0 + \beta_1 * weight + \beta_2 * sleeve + \beta_3 * foot + \epsilon_i, \epsilon_i \sim N(0, \sigma^2).$

```
library(rjags)
army = read.csv("army-physical.csv", header=T, sep=",")
lm.out = lm(height~weight + sleeve + foot, data=army)
summary(lm.out)
plot(lm.out)
modelstring="
  model {
    for (i in 1:n) {
      y[i] ~ dnorm(mu[i], tau)
      mu[i] <- beta0 + inprod(beta[], x[i,])
    }
    tau <- 1/sigma2
    log(sigma2) <- 2*log.sigma
    log(sigma) <- log.sigma
    log.sigma ~ dunif(L2, U2)
    beta0 ~ dunif(L1, U1)
    for (j in 1:p) {
      beta[j] ~ dunif(L1, U1)
    }
    for (l in 1:k) {
      pred[l] <- beta0 + inprod(beta[], px[l,])
    }
  }
"
```

```
L1 = -100; U2 = 100
L2 = -100; U2 = 100
n = dim(army)[1]; p = 3
x = army[,c("weight", "sleeve", "foot")]
px = x[c(3,10),]
k = dim(px)[1]
datalist = list('y'= army$height, 'x'=x, 'px'=px, 'n'=n, 'p'=p,
               'k'=k, 'L1'=L1, 'L2'=L2, 'U1'=U1, 'U2'=U2)
initlist = list('beta'=coef(lm.out)[2:4], 'beta0'=coef(lm.out)[1])

jags <- jags.model(textConnection(modelstring), data = datalist,
                  inits=initlist, n.chains = 1, n.adapt = 100)
post <- coda.samples(jags, c('beta0', 'beta', 'sigma', 'pred'),
                    n.iter=10000, thin=5, burnin=2000)

library(dplyr)
library(coda)
library(ggmcmc)
post %>% summary
post[,c("beta[1]", "sigma")] %>% ggs %>% ggs_density
post[,c("beta[1]", "sigma")] %>% ggs %>% ggs_traceplot
post[,c("beta[1]", "sigma")] %>% ggs %>% ggs_autocorrelation
```

02 로버스트 선형회귀모형

로버스트 선형회귀모형

□ 모형

$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \epsilon_i, \epsilon_i \stackrel{iid}{\sim} t_v(0, \sigma^2), i = 1, 2, \dots, n$
 p 는 변수의 개수이고, $q = p + 1$ 이다.

- ★ 로버스트 모형은 이상치에 둔감한 모형을 말한다.
- ★ 오차의 분포로 꼬리가 정규분포 보다 두꺼운 t -분포나 라플라스 분포를 쓴다.
- ★ t -분포의 밀도함수

$$X \sim t_v(\mu, \sigma^2) \text{ 일 때,}$$

$$f(x | \mu, \sigma^2, v) = \frac{\Gamma(\frac{v+1}{2})}{\sqrt{v\pi\sigma^2} \Gamma(\frac{v}{2})} \cdot \left(1 + \frac{(x - \mu)^2}{v\sigma^2}\right)^{-(v+1)/2}, x \in \mathbb{R}.$$

$$\mathbb{E}X = \mu, v > 1 \quad \text{Var}(X) = \frac{v}{v-2} \sigma^2, v > 2.$$

로버스트 선형회귀모형: 코드

□모형 $y_i = \beta_0 + \beta_1 * \text{weight} + \beta_2 * \text{sleeve} + \beta_3 * \text{foot} + \epsilon_i, \epsilon_i \sim t_v(0, \sigma^2).$

```
library(rjags)
army = read.csv("army-physical.csv", header=T, sep=",")
lm.out = lm(height~weight + sleeve + foot, data=army)
summary(lm.out)
plot(lm.out)
```

```
modelstring="
model {
  for (i in 1:n) {
    y[i] ~ dt(mu[i], tau, dof)
    mu[i] <- beta0 + inprod(beta[], x[i,])
  }
  dof <- 5
  tau <- 1/sigma2
  log(sigma2) <- 2*log.sigma
  log(sigma) <- log.sigma
  log.sigma ~ dunif(L2, U2)
  beta0 ~ dunif(L1, U1)
  for (j in 1:p) {
    beta[j] ~ dunif(L1, U1)
  }
  for (l in 1:k) {
    pred[l] <- beta0 + inprod(beta[], px[l,])
  }
}
```

```
L1 = -100; U2 = 100
L2 = -100; U2 = 100
n = dim(army)[1]; p = 3
x = army[,c("weight", "sleeve", "foot")]
px = x[c(3,10),]
k = dim(px)[1]
datalist = list('y'= army$height, 'x'=x, 'px'=px, 'n'=n, 'p'=p,
'k'=k, 'L1'=L1, 'L2'=L2, 'U1'=U1, 'U2'=U2)
initlist = list('beta'=coef(lm.out)[2:4], 'beta0'=coef(lm.out)[1])
```

```
jags <- jags.model(textConnection(modelstring), data = datalist,
inits=initlist, n.chains = 1, n.adapt = 100)
post <- coda.samples(jags, c('beta0', 'beta', 'sigma', 'pred'),
n.iter=10000, thin=5, burnin=2000)
```

```
library(dplyr)
library(coda)
library(ggmcmc)
post %>% summary
post[,c("beta[1]", "sigma")] %>% ggs %>% ggs_density
post[,c("beta[1]", "sigma")] %>% ggs %>% ggs_traceplot
post[,c("beta[1]", "sigma")] %>% ggs %>% ggs_autocorrelation
```

참고문헌

1. Hoff, P. D. (2009). A first course in Bayesian statistical methods의 9장.
Springer Science & Business Media.



수고하셨습니다.

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감사합니다.

